Array transfer

Given data in the form of an array *Arr* of size *N*. You have to transfer a subsequence of size *M* to the client.

The time taken to transfer a subsequence of size *M* is the product of the *first* and *last* element of the subsequence.

As you do not want your client to wait, you want to minimize the time taken to transfer the subsequence of size *M*.

What is the minimum time the client has to wait for a subsequence of size *M*?

Function description

Complete the function *solve()*. This function takes the following *3* parameters and returns the required answer:

- *N:* Represents an integer denoting the size of the given array *Arr*
- *M:* Represents an integer denoting the size of the subsequence
- *Arr:* Represents an integer array of size *N* representing the given array

an integer array of size N repr
given array

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code.

- The first line of input contains an integer T denoting the number of test cases.
- For each test case:
  - The first line contains an integer N.
  - The second line contains an integer M.
  - The third lines contain N space-separated integers each denoting the array Arr.

Output format

For each test case in a new line, print an integer representing the minimum time.

Constraints

$1 \leq T \leq 10$

$2 \leq N \leq 10^5$

$2 \leq M \leq N$

8 9 10 11 12 13 14 15 16 17 18 19

Explanation

The first line contains the number of test cases, $T = 2$.

For test case 1

Assumptions

- $N = 7$
- $M = 5$
- $Arr = [2, 3, 4, 1, 3, 3, 9]$

Approach

- The subsequence of size 5 are $[2, 3, 4, 1, 3]$, $[2, 3, 4, 1, 9]$, $[2, 4, 1, 3, 3]$, $[2, 4, 1, 3, 9]$, $[3, 4, 1, 3, 9]$, and so on.
- And among all those subsequences the minimum time taken will be from the subsequence $[2, 3, 4, 1, 3]$ and is 6

Hence, the answer is 6.

For test case 2

Assumptions

- $N = 5$
- $M = 5$
- $Arr = [3, 3, 3, 3, 3]$

Approach

Constraints

$1 \leq T \leq 10$

$2 \leq N \leq 10^5$

$2 \leq M \leq N$

$1 \leq Arr[i] \leq 10^9 \;, 1 \leq i \leq N$

Sample input

```
2
7
5
2 3 4 1 3 3 9
5
5
3 3 3 3 3
```

Sample output

```
6
9
```

Explanation

The first line contains the number of test cases, $T = 2$.

For test case *1*

9
10
11
12
13
14
15
16
17
18
19

For test case 2

Assumptions

- $N = 5$
- $M = 5$
- $Arr = [3, 3, 3, 3, 3]$

Approach

- The subsequence of size 5 is $[3, 3, 3, 3, 3]$.
- There is only one subsequence so the minimum time is 9

Hence, the answer is 9.

> ⓘ The following test cases are the actual test cases of this question that may be used to evaluate your submission.

Sample input 1

```
2
12
2
32644 4705 17065 11108 10220
3
3
17434 3658 7313
```

Sample output 1

```
3373485
127494842
```

8
9
10
11
12
13
14
15
16
17
18
19

8 9 10 11 12 13 14 15 16 17 18 19

```
24
8
2224 2611 30328 15982 25259 :
6
5
4572 516 10823 20855 7530 20(
19
16
6919 31189 24086 25729 22568
```

```
6451824
10663140
2933656
91620
1556276
```

View more

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 5.0 sec(s) for each input file
Memory Limit: 256 MB
Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, Python 3.8, Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic missions

Auto-complete ready!

```
}

int main() {

    ios::sync_with_stdio(sync: 0);
    cin.tie(tiestr: 0);
    int T;
    cin >> T;
    for(int t_i = 0; t_i < T; t_i++)
    {
        int N;
        cin >> N;
        int M;
        cin >> M;
        vector<int> Arr(n: N);
        for(int i_Arr = 0; i_Arr < N; i_Arr++)
        {
            cin >> Arr[i_Arr];
        }

        long long out_;
        out_ = solve(N, M, Arr);
        cout << out_;
        cout << "\n";
    }
}
```

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8, Racket,

Test against custom input

Custom input populated

Search

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... an integer array of size N repr...
given array

Input format for custom testing

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- For each test case:
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  - The third lines contain N space-separated integers each denoting the array Arr.

Output format

For each test case in a new line, print an integer representing the minimum time.

Constraints

$1 \leq T \leq 10$

$2 \leq N \leq 10^5$

$2 \leq M \leq N$

9
10
11
12
13
14
15
16
17
18
19

Constraints

$1 \le T \le 10$

$2 \le N \le 10^5$

$2 \le M \le N$

$1 \le Arr[i] \le 10^9$, $1 \le i \le N$

Sample input

```
2
7
5
2 3 4 1 3 3 9
5
5
3 3 3 3 3
```

Sample output

```
6
9
```

Explanation

The first line contains the number of test cases, $T = 2$.

For test case *1*

8 9 10 11 12 13 14 15 16 17 18 19

Explanation

The first line contains the number of test cases, $T = 2$.

For test case 1

Assumptions

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Approach

- The subsequence of size 5 are $[2, 3, 4, 1, 3]$, $[2, 3, 4, 1, 9]$, $[2, 4, 1, 3, 3]$, $[2, 4, 1, 3, 9]$, $[3, 4, 1, 3, 9]$, and so on.
- And among all those subsequences the minimum time taken will be from the subsequence $[2, 3, 4, 1, 3]$ and is 6

Hence, the answer is 6.

For test case 2

Assumptions

- $N = 5$
- $M = 5$
- $Arr = [3, 3, 3, 3, 3]$

Approach

Question 19 | Max. score: 50.00

Maximum displacement

Given a 1D lane, you can move one step ahead (denoted by S) or move one step back (denoted by R). You are given a string of commands and an integer N.

Calculate the maximum displacement that you can achieve from the starting point after changing exactly N commands in the string and executing them.

Function description

Complete the *Game()* function. This function takes the following 2 parameters and returns the required answer:

- *Str:* Represents the string containing the commands
- *N:* Represents the number of commands that must be changed

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code.

- The first line contains the string *Str* which represents the

New Submission | All Submissions

Auto-complete ready! | Save | C++17 (g++ 10.3.0)

```cpp
#include<bits/stdc++.h>
using namespace std;

int Game (string Str, int N) {
    // Write your code here

}

int main() {

    ios::sync_with_stdio(sync: 0);
    cin.tie(tiestr: 0);
    string Str;
    cin >> Str;
    int N;
    cin >> N;

    int out_;
    out_ = Game(Str, N);
    cout << out_;
}
```

1:1 vscode

Test against custom input

Custom input populated

Compile & Test code | Submit code

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code.

- The first line contains the string *Str* which represents the string that contains the commands.
- The second line contains an integer *N* which represents the number of commands that must be changed.

Output format

Print the maximum displacement that you can achieve from the starting point after changing exactly *N* commands in the string and executing all the commands of that string. One command can be changed several times.

Constraints

$1 \leq |Str| \leq 100$

$1 \leq N \leq 50$

Sample input

```
SR
1
```

Sample output

```
2
```

Question 1 | Max. score: 50.00

Alice and array partitioning

You are given an array, A of length N, and also a single integer K. Your task is to split the array A into K non-overlapping, non-empty subarrays. For each of the subarrays, you calculate the sum of the elements in it. Let us denote these sums as $S_1, S_2, S_3, \ldots, S_K$. Where S_i denotes the sum of the elements in the i^{th} subarray from left.

Let $G = GCD(S_1, S_2, S_3, \ldots, S_K)$.

Task

Print the maximum value of G.

Notes

- GCD of x and y is the largest integer that divides both x and y.
- GCD of a single element is the element itself.
- A subarray is a set of *contiguous* elements in the array.
- Each element of the array should belong to *exactly* one subarray.

Example

Assumptions

New Submission | All Submissions

Auto-complete ready! | Save | C++ (g++ 10.3.0)

```
//          for(int i=0;i<N;i++){
//              sum+=A[i];
//          }
//          return sum;
//      }
// //    if(K==2){

// //    }
// //    cout<<gcd(6,12,27,3)<<endl;
//      // vector<vector<int>> v;
//      vector<int> v;
//      for(int i=0;i<N-K+1;i++){
//          int a = sum(A,i,)
//      }

//    return 0;
    int ans=0;
    if(N==5 && K==3){

    }
}

int main() {

```

49:15 vscode

Test against custom input

Custom input populated

Compile & Test code | Submit code

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7 / 19 Completed

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Question 9

Max score: 6.00

In Data Structures and Algorithms, which of the following statements about the **Trie** data structure are correct:

1. A key can be searched with O(M+N) time complexity by using this data structure where 'M' is the maximum string length and 'N' is the number of keys in the trie.
2. A Trie node field isEndOfWord is used to distinguish the node as the end of the word node.
3. Every character of the input key is inserted as an individual Trie node when a node is being inserted into the Trie.
4. If the isEndofWord field of the last node is true while searching for a key, then the key exists in the trie.

- 1, 2, and 3
- 2, 3, and 4
- 1, 2, and 4
- 1, 3, and 4

Reset Answer

< Previous Question

Search

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You have successfully completed the Netradyne Qualifying Test - BITS Goa 2024 - ND0120 and we have received all your submissions.

The summary of your test is as follows:

- **Multiple Choice Questions questions**
    - Total number of questions: 17
    - Submitted: 14
- **Programming questions**
    - Total number of questions: 2
    - Submitted: 2

If you have any queries, send an e-mail to mohan.t@netradyne.com

Regards,

Team Netradyne

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End test

Question 1

Max. score[redacted]80.00

New Submission   All Submissions

Make Them Equal

You are given an array *A* of size *N*.

You can perform the following operation on the array.

- First, choose any integer *X* that should be a power of *2*.
- Choose any index *i* from *0* to *N-1* and replace *A[i]* with *A[i]* | *X*. Here | denotes OR operation.

Then you are given *Q* queries.

Each query contains two integers *L* and *R*.

You have to find the minimum operation needed to make all the elements in the subarray from the *Lth* index to the *Rth* index equal.

Notes

Auto-complete ready!   Save   C++17 (g++ 10.3.0)

```
#include<bits/stdc++.h>
using namespace std;

vector<int> Solve (int N, vector<int> A, int Q, vector<vector<int> > query) {
    // Write your code here

}

int main() {

    ios::sync_with_stdio(0);
    cin.tie(0);
    int T;
    cin >> T;
    for(int t_i = 0; t_i < T; t_i++)
    {
        int N;
```

1:1 vscode

Test against custom input

Compile & Test code   Submit code